Unstructured Database - ASSIGNMENT - 6

Schema - :

{Schema { emp_id, name, department, designation, city, salary, experience }

Inserting their data :

```
test> db.employees.insertMany([
|   { emp_id: 101, name: "Amit Sharma", department: "IT", designation: "Developer", city: "Delhi", salary: 60000, experiexperience: 3 },
|   { emp_id: 102, name: "Riya Sen", department: "HR", designation: "Manager", city: "Kolkata", salary: 45000, experiencexperience: 5 },
|   { emp_id: 103, name: "Rahul Das", department: "Finance", designation: "Analyst", city: "Mumbai", salary: 70000, expeexperience: 6 },
|   { emp_id: 104, name: "Sneha Roy", department: "IT", designation: "Developer", city: "Kolkata", salary: 80000, experiexperience: 7 },
|   { emp_id: 105, name: "Arjun Kapoor", department: "Marketing", designation: "Executive", city: "Delhi", salary: 50000, experience: 2 },
|   { emp_id: 106, name: "Neha Gupta", department: "HR", designation: "Executive", city: "Pune", salary: 55000, experienexperience: 4 },
|   { emp_id: 107, name: "Karan Mehta", department: "IT", designation: "Tester", city: "Delhi", salary: 48000, experiencexperience: 1 },
|   { emp_id: 108, name: "Priya Singh", department: "Finance", designation: "Manager", city: "Kolkata", salary: 75000, eexperience: 8 },
|   { emp_id: 109, name: "Vikas Yadav", department: "IT", designation: "Developer", city: "Mumbai", salary: 65000, experexperience: 5 },
|   { emp_id: 110, name: "Anjali Verma", department: "Marketing", designation: "Manager", city: "Delhi", salary: 72000, experience: 9 }
| ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69db3b015dbbb6396a3682d1'),
    '1': ObjectId('69db3b015dbbb6396a3682d2'),
    '2': ObjectId('69db3b015dbbb6396a3682d3'),
    '3': ObjectId('69db3b015dbbb6396a3682d4'),
    '4': ObjectId('69db3b015dbbb6396a3682d5'),
    '5': ObjectId('69db3b015dbbb6396a3682d6'),
    '6': ObjectId('69db3b015dbbb6396a3682d7'),
    '7': ObjectId('69db3b015dbbb6396a3682d8'),
    '8': ObjectId('69db3b015dbbb6396a3682d9'),
    '9': ObjectId('69db3b015dbbb6396a3682da')
  }
}
test>
```

1. **Display employee name and salary only.**

```
test> db.employees.find({}, { name: 1, salary: 1, _id: 0 });
[
  { name: 'Amit Sharma', salary: 60000 },
  { name: 'Riya Sen', salary: 45000 },
  { name: 'Rahul Das', salary: 70000 },
  { name: 'Sneha Roy', salary: 80000 },
  { name: 'Arjun Kapoor', salary: 50000 },
  { name: 'Neha Gupta', salary: 55000 },
  { name: 'Karan Mehta', salary: 48000 },
  { name: 'Priya Singh', salary: 75000 },
  { name: 'Vikas Yadav', salary: 65000 },
  { name: 'Anjali Verma', salary: 72000 }
]
test>
```

2. Find employees working in HR department.

```
test> db.employees.find({ department: "HR" });
[
  {
    _id: ObjectId('69db3b015dbbb6396a3682d2'),
    emp_id: 102,
    name: 'Riya Sen',
    department: 'HR',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 45000,
    experience: 5
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d6'),
    emp_id: 106,
    name: 'Neha Gupta',
    department: 'HR',
    designation: 'Executive',
    city: 'Pune',
    salary: 55000,
    experience: 4
  }
]
test>
```

3. Find employees whose salary is between 50,000 and 80,000.

```
    city: 'Pune',
    salary: 55000,
    experience: 4
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d8'),
    emp_id: 108,
    name: 'Priya Singh',
    department: 'Finance',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d9'),
    emp_id: 109,
    name: 'Vikas Yadav',
    department: 'IT',
    designation: 'Developer',
    city: 'Mumbai',
    salary: 65000,
    experience: 5
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682da'),
    emp_id: 110,
    name: 'Anjali Verma',
    department: 'Marketing',
    designation: 'Manager',
    city: 'Delhi',
    salary: 72000,
    experience: 9
  }
]
test>
```

```
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d3'),
    emp_id: 103,
    name: 'Rahul Das',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 70000,
    experience: 6
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d4'),
    emp_id: 104,
    name: 'Sneha Roy',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 80000,
    experience: 7
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d5'),
    emp_id: 105,
    name: 'Arjun Kapoor',
    department: 'Marketing',
    designation: 'Executive',
    city: 'Delhi',
    salary: 50000,
    experience: 2
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d6'),
    emp_id: 106,
    name: 'Neha Gupta',
    department: 'HR',
    designation: 'Executive',
    city: 'Pune',
    salary: 55000,
    experience: 4
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d8'),
    emp_id: 108,
    name: 'Priya Singh',
    department: 'Finance',
    designation: 'Manager',
    city: 'Kolkata',
```

```
test> db.employees.find({
|   salary: { $gte: 50000, $lte: 80000 }
| });
[
  {
    _id: ObjectId('69db3b015dbbb6396a3682d1'),
    emp_id: 101,
    name: 'Amit Sharma',
    department: 'IT',
    designation: 'Developer',
    city: 'Delhi',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d3'),
    emp_id: 103,
    name: 'Rahul Das',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 70000,
    experience: 6
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d4'),
    emp_id: 104,
    name: 'Sneha Roy',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 80000,
    experience: 7
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d5'),

test> db.employees.find({
|   salary: { $gte: 50000, $lte: 80000 }
| });
[
  {
    _id: ObjectId('69db3b015dbbb6396a3682d1'),
    emp_id: 101,
    name: 'Amit Sharma',
    department: 'IT',
    designation: 'Developer',
    city: 'Delhi',
    salary: 60000,
    experience: 3
```

4. Display employees from Kolkata city.

```
test> db.employees.find({ city: "Kolkata" });
[
  {
    _id: ObjectId('69db3b015dbbb6396a3682d2'),
    emp_id: 102,
    name: 'Riya Sen',
    department: 'HR',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 45000,
    experience: 5
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d4'),
    emp_id: 104,
    name: 'Sneha Roy',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 80000,
    experience: 7
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d8'),
    emp_id: 108,
    name: 'Priya Singh',
    department: 'Finance',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 75000,
    experience: 8
  }
]
test>
```

5. Sort employees by experience in ascending order.

```
test> db.employees.find().sort({ experience: 1 });
[
  {
    _id: ObjectId('69db3b015dbbb6396a3682d7'),
    emp_id: 107,
    name: 'Karan Mehta',
    department: 'IT',
    designation: 'Tester',
    city: 'Delhi',
    salary: 48000,
    experience: 1
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d5'),
    emp_id: 105,
    name: 'Arjun Kapoor',
    department: 'Marketing',
    designation: 'Executive',
    city: 'Delhi',
    salary: 50000,
    experience: 2
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d1'),
    emp_id: 101,
    name: 'Amit Sharma',
    department: 'IT',
    designation: 'Developer',
    city: 'Delhi',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d6'),
    emp_id: 106,
    name: 'Neha Gupta',
    department: 'HR',
    designation: 'Executive',
    city: 'Pune',
    salary: 55000,
    experience: 4
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d2'),
    emp_id: 102,
    name: 'Riya Sen',
    department: 'HR',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 45000,
```

```
    city: 'Kolkata',
    salary: 45000,
    experience: 5
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d9'),
    emp_id: 109,
    name: 'Vikas Yadav',
    department: 'IT',
    designation: 'Developer',
    city: 'Mumbai',
    salary: 65000,
    experience: 5
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d3'),
    emp_id: 103,
    name: 'Rahul Das',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 70000,
    experience: 6
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d4'),
    emp_id: 104,
    name: 'Sneha Roy',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 80000,
    experience: 7
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d8'),
    emp_id: 108,
    name: 'Priya Singh',
    department: 'Finance',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682da'),
    emp_id: 110,
    name: 'Anjali Verma',
    department: 'Marketing',
    designation: 'Manager',
```

```
    _id: ObjectId('69db3b015dbbb6396a3682d3'),
    emp_id: 103,
    name: 'Rahul Das',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 70000,
    experience: 6
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d4'),
    emp_id: 104,
    name: 'Sneha Roy',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 80000,
    experience: 7
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d8'),
    emp_id: 108,
    name: 'Priya Singh',
    department: 'Finance',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682da'),
    emp_id: 110,
    name: 'Anjali Verma',
    department: 'Marketing',
    designation: 'Manager',
    city: 'Delhi',
    salary: 72000,
    experience: 9
  }
]
test>
```

6. Display top 5 most experienced employees.

```
    _id: ObjectId('69db3b015dbbb6396a3682d9'),

test> db.employees.find().sort({ experience: -1 }).limit(5);
[
  {
    _id: ObjectId('69db3b015dbbb6396a3682da'),
    emp_id: 110,
    name: 'Anjali Verma',
    department: 'Marketing',
    designation: 'Manager',
    city: 'Delhi',
    salary: 72000,
    experience: 9
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d8'),
    emp_id: 108,
    name: 'Priya Singh',
    department: 'Finance',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d4'),
    emp_id: 104,
    name: 'Sneha Roy',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 80000,
    experience: 7
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d3'),
    emp_id: 103,
    name: 'Rahul Das',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 70000,
    experience: 6
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d9'),
    emp_id: 109,
    name: 'Vikas Yadav',
    department: 'IT',
    designation: 'Developer',
```

7. Update salary of employee with emp_id 102 to 48,000.

```
test> db.employees.updateOne(
|   { emp_id: 102 },
|   { $set: { salary: 48000 } }
| );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test>
```

8. Increase salary of all IT employees by 10,000.

```
test> db.employees.updateMany(
|   { department: "IT" },
|   { $inc: { salary: 10000 } }
| );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
test>
```

9. **Update designation to “Senior Developer” for IT employees having experience more than 5 years.**

```
test> db.employees.updateMany(
|   { department: "IT", experience: { $gt: 5 } },
|   { $set: { designation: "Senior Developer" } }
| );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test>
```

10. **Delete employees whose experience is less than 2 years.**

```
test> db.employees.deleteMany({
|   experience: { $lt: 2 }
| });
{ acknowledged: true, deletedCount: 1 }
test>
```

11. **Display employees who are not from Delhi.**

```
  {
    _id: ObjectId('69db3b015dbbb6396a3682d6'),
    emp_id: 106,
    name: 'Neha Gupta',
    department: 'HR',
    designation: 'Executive',
    city: 'Pune',
    salary: 55000,
    experience: 4
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d8'),
    emp_id: 108,
    name: 'Priya Singh',
    department: 'Finance',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d9'),
    emp_id: 109,
    name: 'Vikas Yadav',
    department: 'IT',
    designation: 'Developer',
    city: 'Mumbai',
    salary: 75000,
    experience: 5
  }
]
test>
```

```
test> db.employees.find({
|   city: { $ne: "Delhi" }
| });
[
  {
    _id: ObjectId('69db3b015dbbb6396a3682d2'),
    emp_id: 102,
    name: 'Riya Sen',
    department: 'HR',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 45000,
    experience: 5
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d3'),
    emp_id: 103,
    name: 'Rahul Das',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 70000,
    experience: 6
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d4'),
    emp_id: 104,
    name: 'Sneha Roy',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Kolkata',
    salary: 90000,
    experience: 7
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d6'),
    emp_id: 106,
    name: 'Neha Gupta',
    department: 'HR',
    designation: 'Executive',
    city: 'Pune',
    salary: 55000,
    experience: 4
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d8'),
    emp_id: 108,
    name: 'Priya Singh',
```

12. Find employees whose designation is Developer and salary greater than 60,000.

```
test> db.employees.find({
|   designation: "Developer",
|   salary: { $gt: 60000 }
| });
[
  {
    _id: ObjectId('69db3b015dbbb6396a3682d1'),
    emp_id: 101,
    name: 'Amit Sharma',
    department: 'IT',
    designation: 'Developer',
    city: 'Delhi',
    salary: 70000,
    experience: 3
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d9'),
    emp_id: 109,
    name: 'Vikas Yadav',
    department: 'IT',
    designation: 'Developer',
    city: 'Mumbai',
    salary: 75000,
    experience: 5
  }
]
test>
```

13. Display employees working in IT or Finance department.

```
test> db.employees.find({
|   department: { $in: ["IT", "Finance"] }
| });
[
  {
    _id: ObjectId('69db3b015dbbb6396a3682d1'),
    emp_id: 101,
    name: 'Amit Sharma',
    department: 'IT',
    designation: 'Developer',
    city: 'Delhi',
    salary: 70000,
    experience: 3
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d3'),
    emp_id: 103,
    name: 'Rahul Das',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 70000,
    experience: 6
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d4'),
    emp_id: 104,
    name: 'Sneha Roy',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Kolkata',
    salary: 90000,
    experience: 7
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d8'),
    emp_id: 108,
    name: 'Priya Singh',
    department: 'Finance',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d9'),
    emp_id: 109,
    name: 'Vikas Yadav',
    department: 'IT',
    designation: 'Developer',
```

14. Display employees sorted by department and then by salary.

```
test> db.employees.find().sort({
|   department: 1,
|   salary: 1
| });
[
  {
    _id: ObjectId('69db3b015dbbb6396a3682d3'),
    emp_id: 103,
    name: 'Rahul Das',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 70000,
    experience: 6
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d8'),
    emp_id: 108,
    name: 'Priya Singh',
    department: 'Finance',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d2'),
    emp_id: 102,
    name: 'Riya Sen',
    department: 'HR',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 45000,
    experience: 5
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d6'),
    emp_id: 106,
    name: 'Neha Gupta',
    department: 'HR',
    designation: 'Executive',
    city: 'Pune',
    salary: 55000,
    experience: 4
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d1'),
    emp_id: 101,
    name: 'Amit Sharma',
```

15. Count total number of employees in the company.

```
test> db.employees.countDocuments();
9
test>
```

16. Display employees whose salary is not equal to 50,000

```
test> db.employees.find({
|   salary: { $ne: 50000 }
| });
[
  {
    _id: ObjectId('69db3b015dbbb6396a3682d1'),
    emp_id: 101,
    name: 'Amit Sharma',
    department: 'IT',
    designation: 'Developer',
    city: 'Delhi',
    salary: 70000,
    experience: 3
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d2'),
    emp_id: 102,
    name: 'Riya Sen',
    department: 'HR',
    designation: 'Manager',
    city: 'Kolkata',
    salary: 45000,
    experience: 5
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d3'),
    emp_id: 103,
    name: 'Rahul Das',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 70000,
    experience: 6
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d4'),
    emp_id: 104,
    name: 'Sneha Roy',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Kolkata',
    salary: 90000,
    experience: 7
  },
  {
    _id: ObjectId('69db3b015dbbb6396a3682d6'),
    emp_id: 106,
    name: 'Neha Gupta',
    department: 'HR',
    designation: 'Executive',
```